

Muhammad Taqi Abbas

Passport: [redacted] | **Place of birth:** Lahore, Pakistan | **Phone:** [redacted] | **Email address:** muhammadtaqi811@gmail.com | **LinkedIn:** www.linkedin.com/in/muhammad-taqi-5221182b3 | **Address:** [redacted]

ABOUT ME

Biomedical Engineering researcher (CGPA 3.81/4.00) specializing in the convergence of artificial intelligence, biomedical signal processing, medical imaging, and assistive technologies. Experienced in developing intelligent healthcare systems using machine learning and deep learning for physiological signal analysis, including CNN-based lung nodule malignancy detection and EMG-based hand gesture recognition for assistive hand exoskeletons. My research combines signal processing, machine learning, embedded systems, and human–machine interaction to develop data-driven solutions for rehabilitation and clinical applications. Currently focused on advancing AI-enabled biomedical systems, neural engineering, wearable sensing, Robotics and intelligent assistive technologies, with particular interest in Physiological Signals based interfaces, medical imaging, neurorehabilitation, and robotics. Published research in biomedical AI and actively contributing to multidisciplinary projects aimed at translating intelligent sensing and computational methods into practical healthcare technologies.

EDUCATION & TRAINING

2021/09/21 - 2025/07/04 - LAHORE, PAKISTAN

BSC BIOMEDICAL ENGINEERING- RIPHAH INTERNATIONAL UNIVERSITY

Biomedical Signal Processing, Medical Image Processing, Artificial Intelligence, Biomedical Control Systems

Final grade: CGPA:3.81/4.00 | **Number of credits:** 139 | **Thesis:** Real-time EMG Pattern Recognition-Based Assistive Hand Exoskeleton for Cervical Spinal Cord Injury Patients

WORK EXPERIENCE

2026/01/05 - CURRENT - LAHORE, PAKISTAN

BIOMEDICAL ENGINEERSHALAMAR HOSPITAL

- Assisted in the installation, calibration, and preventive maintenance of biomedical equipment, including patient monitors and ventilators.
- Supported the diagnosis and troubleshooting of medical devices, ensuring timely repair and minimal equipment downtime in critical care units.

Business or sector Human health and social work activities | **Department:** Biomedical Services

2025/08/05 - 2026/01/05 - LAHORE, PAKISTAN

SALES & SERVICE ENGINEERVERTEX MEDICAL PVT LTD

- Delivered product demonstrations and user training for healthcare staff
- Provided technical support and resolved client issues promptly.
- Assisted the sales team with technical presentations and client workshops

Business or sector Human health and social work activities | **Department:** Biomedical Services

2024/01/15 - 2025/07/14 - LAHORE, PAKISTAN

RESEARCH ASSISTANTRIPHAH INTERNATIONAL UNIVERSITY

- My work includes helping with biomedical signal processing tasks, sensor integration, data collection,

embedded systems programming, and troubleshooting hardware issues.

- I also support lab management activities, assist students during experiments, help with literature reviews and documentation, and contribute to projects related to wearable healthcare devices, IoT-based monitoring systems, and AI applications in healthcare.

Business or sector Education | **Department:** Riphah College Of Science & Technology

2019/02/25 - CURRENT - LAHORE, PAKISTAN

INSTRUCTOR ONLINE & HOME

- Teach STEM concepts, including science, technology, engineering, and mathematics, through online sessions.
- Explain technical concepts using practical examples and structured learning activities.
- Prepare lessons and adapt teaching methods to students' learning needs.
- Support students in developing problem-solving, analytical thinking, and technical understanding.

Business or sector Education

SKILLS

Programming & Data Analysis
C++ | Python (computer programming) | MATLAB

Biomedical Sensors & Instrumentation
EMG | ECG | ESP32 | ARDUINO | Temperature Sensors | SPO2 | PROTEUS

Biomedical Design
COMSOL Multiphysics | Solidworks | AUTOCAD

Software & Office Tools
Microsoft Office | MULTISIM | Spreadsheets

LANGUAGE SKILLS

Mother tongue(s): **URDU**

Other language(s): **ENGLISH: PROFESSIONAL WORKING PROFICIENCY**

PUBLICATIONS

Detecting Lung Nodule Malignancy Using CNN-Based Deep Learning Techniques 2025

Mishaal Khawar, Muhammad Taqi Abbas, Faizan Ahmad, Komal Tariq, & Saba Anwer Institute of Electrical and Electronics Engineers(IEEE) 3rd International Conference in Electrical, Control, and Telecommunication Engineering.

The Impact of Smog on Public Health and Antimicrobial Resistance in Pakistan. 2025

Muhammad Taqi Abbas, Mishaal Khawar, & Zainab Munir. MDPI Proceedings of the 7th International Electronic Conference on Atmospheric Sciences

Immersive Technologies in Healthcare: Virtual Reality's Role in Antibiotic Resistance 2025

Mishaal Khawar, Hadiya Malik, Muhammad Taqi Abbas, & Alizay Ashfaq MDPI. Proceedings of the 4th International Electronic Conference on Antibiotics.

Bio-absorbable Antibiotic Beads for Vascular Graft Infection Management 2025

Mishaal Khawar, Muhammad Taqi Abbas, & Alizay Ashfaq MDPI Proceedings of the 4th International Electronic Conference on Antibiotics

CONFERENCES & SEMINARS

2025/06/11 - 2025/06/14 Riphah International University, Lahore, Pakistan

Hyperelastic Model-Based Mechanical Assessment of Corrugated and Non-Corrugated Thermoplastic Polyurethane Grafts Delivered an oral presentation at the International Conference on Emerging Innovations in Science, Engineering & Intelligent Technologies (ICSEIT 2025).

2025/05/13 - 2025/05/16 Riphah International University, Lahore, Pakistan

Orthotic Management of Charcot-Marie-Tooth Disease: Impact on Patient Outcomes Delivered an oral presentation at the International Conference on Emerging Innovations in Science, Engineering & Intelligent Technologies (ICSEIT 2025).

2025/02/11 - 2025/02/13 Lahore University Of Management Sciences(LUMS), Lahore, Pakistan

From Goo to Glow: Transforming Wearable Bioelectronics Using Electro-Conductive Hydrogels Presented at The Particle by Lahore University of Management Sciences (LUMS).

2025/01/06 - 2025/01/08 University of Lahore,Pakistan

Detecting Lung Nodule Malignancy Using CNN-Based Deep Learning Techniques Presented research Paper at the 3rd International Conference on Emerging Trends in Electrical, Control and Telecommunication Engineering (ETECTE 2024).

2024/06/06 - 2024/06/08 Riphah International University, Lahore, Pakistan

UV-Pure Guard: A Multi-Mode UV Sterilization System for Inactivation of Airborne Viruses in Healthcare Settings Presented a poster at Open House Exhibition.

2024/03/06 - 2024/03/07 University of Engineering & Technology, Lahore, Pakistan

Using Conductive Hydrogels for Al-Enhanced Wearable Bioelectronics Presented a Research poster at the 7th International Conference on Energy Conservation and Efficiency (ICECE 2024).

PROJECTS

2024/07/05 - 2025/05/07

- Developed a real-time EMG-based assistive hand exoskeleton for gesture recognition and rehabilitation support.
- Applied signal preprocessing and machine learning techniques to classify hand gestures from EMG signals.
- Integrated EMG-based gesture recognition with microcontroller-driven motor actuation to assist hand movements.

2025/01/11 - 2025/02/13
From Goo to Glow: Transforming Wearable Bioelectronics Using Electro Conductive Hydrogels, The Particle by Lahore University of Management Sciences.

2024/01/12 - 2024/04/10
Comparative Analysis of Binary and Multiclass EMG-Based Hand Gesture Recognition

2024/01/04 - 2024/04/11
Portable Care Taker Patient Monitor

2023/10/12 - 2026/12/17
Arrhythmia Detection Using MATLAB

2023/04/05 - 2023/06/07
ECG System Design

HONOURS AND AWARDS

2025/07/14 Riphah International University
Overall Excellence Award

2025/06/18 Riphah International University
Class Representative

2025/03/11 Riphah International University
General Secretary | Society of Engineering and Sciences

2023/10/17 Riphah International University
Talent Scholarship

2024/10/16 Microsoft Coorporation
2nd Runner-up, Microsoft Imagine Cup Pakistan

VOLUNTEERING

2024/10/08 - 2025/01/06 Lahore
Community Health

2023/10/03 - 2024/10/09 Riphah International University, Lahore, Pakistan
Event Volunteer

2022/09/08 - 2023/10/03 Beacon Tutors Pakistan
STEM Instructor

RECOMMENDATIONS

Dr.Tausif Zahid Associate Professor & Director Campus

Senior academic leader responsible for departmental management and academic affairs.

Oversees faculty coordination, educational development, and campus administration.

Available as an academic referee to support my postgraduate study and scholarship applications.

Email tausif.zahid@riphah.edu.pk | **Phone** +92 3000720698

Dr. Jawad Hussain Associate Professor

- Associate Professor at Riphah International University with expertise in biomedical engineering and academic research.
- Provides academic guidance and evaluates my research abilities, technical skills, and academic performance.
- Recommends me for postgraduate studies based on my academic achievements, research potential, and commitment to biomedical engineering.

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